

Mechanisms & Vibrations

- Forced Vibrations

Course outline

- Support motion
- Vibration isolation
- Vibration sensors

Text: Chapter 3

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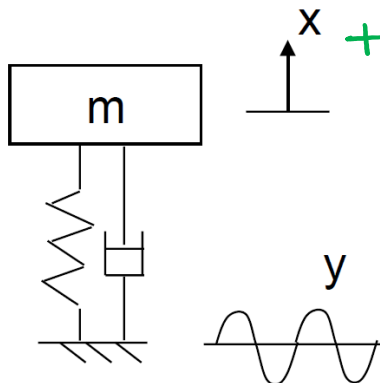
Support motion

In this case, the excitation is a vibrating base.

Examples include:

- Ground vibration
- Aircraft body vibration
- Car on a bumpy road
- Ship on waves
- Earthquake

where is the mass at any given time $x(t)$?

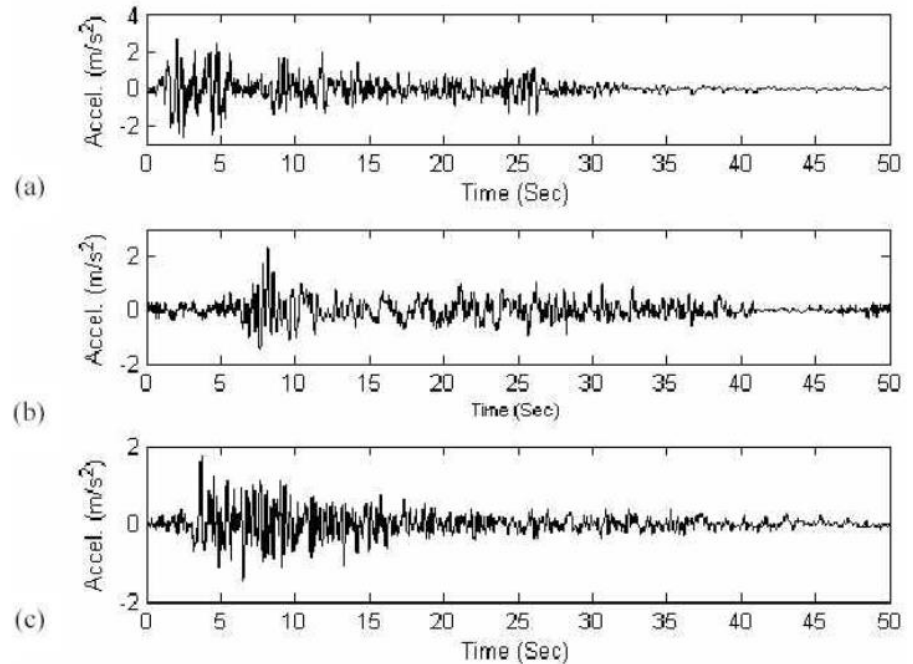
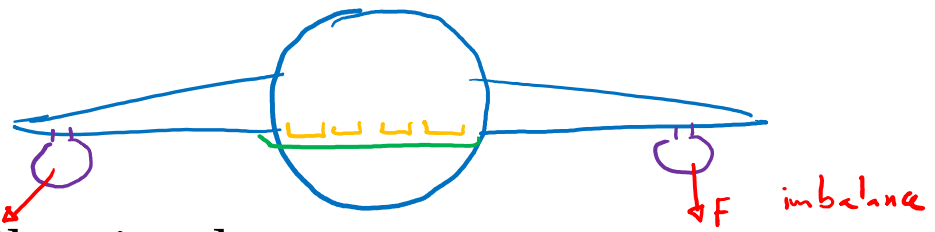


$$c\left(\frac{dx}{dt} - \frac{dy}{dt}\right) + k(x - y)$$

total disp = $x - y$

$$\sum F = ma$$

→ vibrate



Equation of motion

The equation of motion is $\sum F = ma$

$$m d^2x/dt^2 = -k(x-y) - c(dx/dt - dy/dt)$$

Let $m(\ddot{z} + \ddot{y}) = -k(z) - c(\dot{z})$

$$z = x - y \text{ (relative displacement)} \quad z = x - y \rightarrow x = z + y$$

Then

$$m d^2z/dt^2 + c dz/dt + kz = -m d^2y/dt^2 \rightarrow \text{Governing Eq'n}$$

Since the base is vibrating

$$y = Y \sin \omega t \rightarrow m d^2y/dt^2 = -m \omega^2 Y \sin \omega t$$

Hence,

$$m d^2z/dt^2 + c dz/dt + kz = m \omega^2 Y \sin \omega t \quad \text{Final Eq'n}$$

assume

$$\begin{aligned} y(t) &= Y \sin \omega t \\ \dot{y}(t) &= Y \omega \cos \omega t \\ \ddot{y}(t) &= -Y \omega^2 \sin \omega t \end{aligned}$$

This equation is similar to the equation of motion for imbalance induced vibration

Steady-state solution

The steady-state solution is

$$z = Z \sin(\omega t - \phi)$$

Where

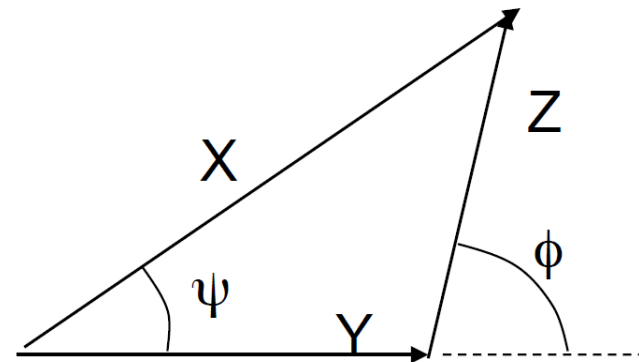
$$\text{Amplitude: } Z = m\omega^2 Y / [(k - m\omega^2)^2 + (c\omega)^2]^{1/2}$$

$$\text{phase: } \phi = \text{Arctan} [c\omega / (k - m\omega^2)]$$

Now the solution of z is converted to x .

Since

$$x = z + y = Z \sin(\omega t - \phi) + Y \sin(\omega t)$$



Steady-state solution

Based on the law of Cosines, then X amplitude is

$$\begin{aligned} X^2 &= Z^2 + Y^2 + 2ZY\cos\phi \\ &= (m\omega^2 Y)^2 / [(k - m\omega^2)^2 + (c\omega)^2] + Y^2 + 2m\omega^2 Y^2 \cos\phi / [(k - m\omega^2)^2 + (c\omega)^2]^{1/2} \\ &= Y^2 [k^2 + (c\omega)^2] / [(k - m\omega^2)^2 + (c\omega)^2] \end{aligned}$$

$$\begin{aligned} |X/Y| &= \sqrt{[k^2 + (c\omega)^2] / [(k - m\omega^2)^2 + (c\omega)^2]} \rightarrow m, c, k \\ &= \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]} \rightarrow \text{dimensionless} \end{aligned}$$

Phase $\tan\psi = Z\sin\phi / (Z\cos\phi + Y)$

Since

$$\sin\phi = (c\omega) / [(k - m\omega^2)^2 + (c\omega)^2]^{1/2}; \cos\phi = (k - m\omega^2) / [(k - m\omega^2)^2 + (c\omega)^2]^{1/2}$$

Then $\tan\psi = Z\sin\phi / (Y + Z\cos\phi) = mc\omega^3 / [(k^2 - km\omega^2 + (c\omega)^2)]$

$$x = X\sin(\omega t - \psi)$$

Steady-state solution-special cases

Final equations for vibnd of a mass (m) under support motion $y(t) = Y \sin \omega t$

$$x = X \sin(\omega t - \psi)$$

$$|X/Y| = \sqrt{[k^2 + (c\omega)^2] / [(k - m\omega^2)^2 + (c\omega)^2]} \quad \leftarrow m, c, k$$

$$= \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]} \quad \leftarrow \text{dimensionless}$$

Phase $\tan \psi = m c \omega^3 / [k^2 - k m \omega^2 + (c\omega)^2] \quad \leftarrow m, c, k$

$$= 2\xi(\omega/\omega_n)^3 / (1 - (\omega/\omega_n)^2 + (2\xi(\omega/\omega_n))^2) \quad \leftarrow \text{dimensionless}$$

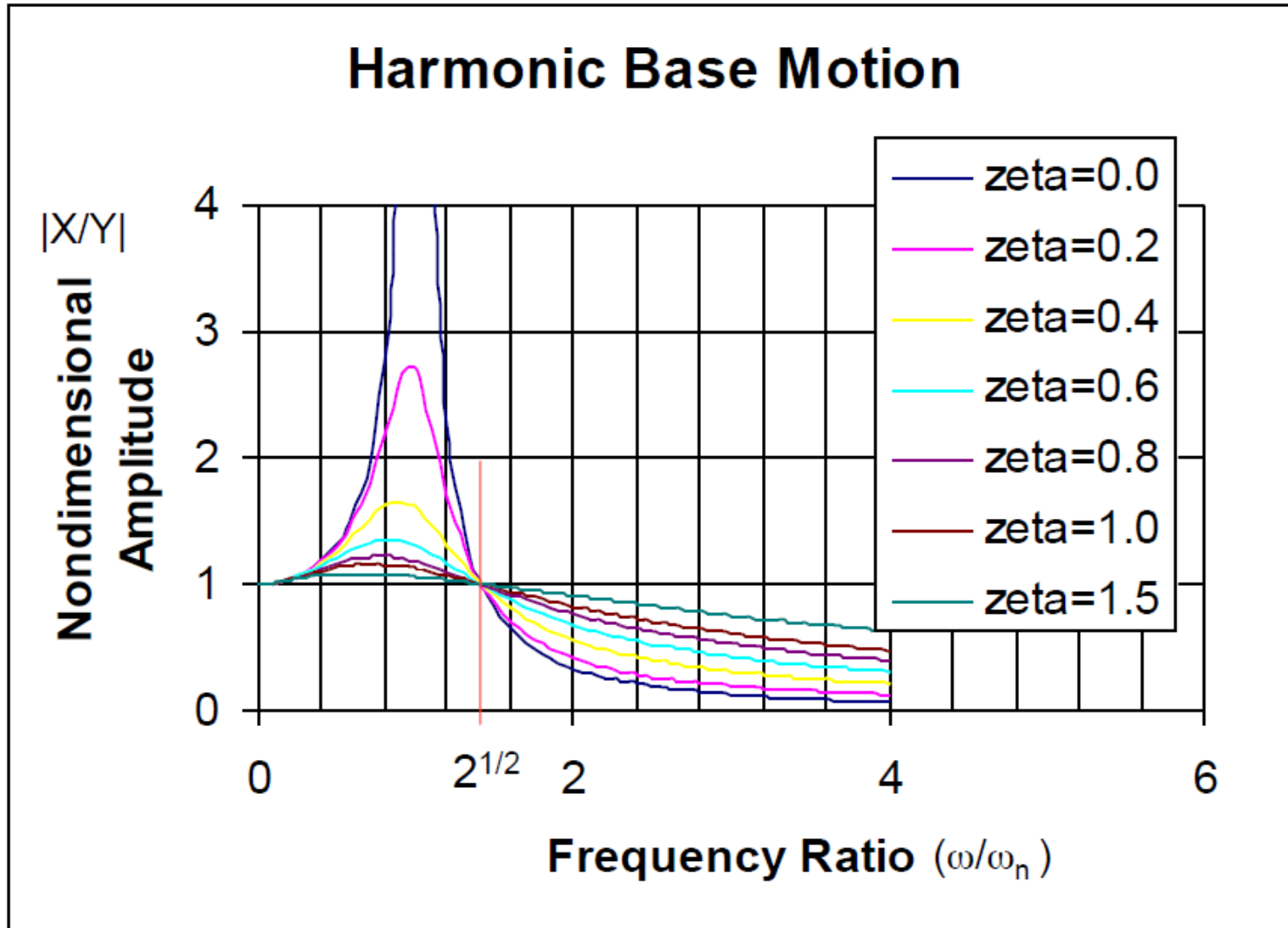
$$\omega/\omega_n = \sqrt{2} \rightarrow |X/Y| = 1$$

$$\omega/\omega_n = 1 \rightarrow \tan \psi = 1/(2\xi)$$

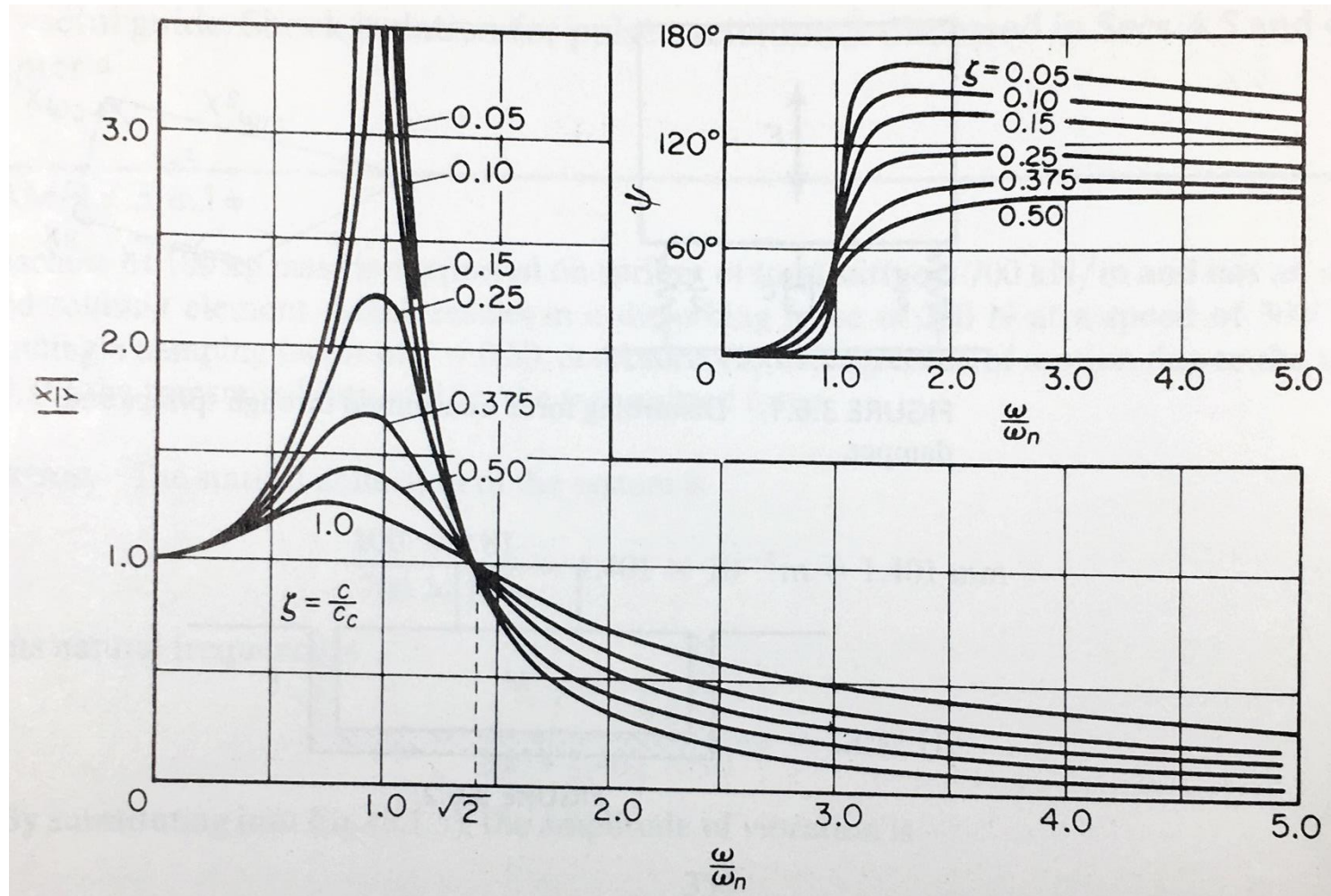
$$\frac{\omega}{\omega_n} = \sqrt{2} \rightarrow \left| \frac{X}{Y} \right| = \sqrt{\frac{1 + (2\xi \frac{\omega}{\omega_n})^2}{(1 - (\frac{\omega}{\omega_n})^2)^2 + (2\xi \frac{\omega}{\omega_n})^2}} = \sqrt{\frac{1 + (2\xi \frac{\omega}{\omega_n})^2}{(\underbrace{1 - (\sqrt{2})^2}_{-1})^2 + (2\xi \frac{\omega}{\omega_n})^2}}$$

$$\rightarrow \left| \frac{X}{Y} \right| = 1$$

Amplitude vs frequency ratio



Phase vs frequency ratio



Superposition method

Original equation

$$m d^2x/dt^2 + c dx/dt + kx = ky + c dy/dt$$

$$Y = Y \sin \omega t, \quad dy/dt = Y \omega \cos \omega t$$

Solution 1

$$m d^2x/dt^2 + c dx/dt + kx = ky = kY \sin \omega t$$

$$x_1 = (F/k) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]^{1/2} \sin(\omega t - \phi)$$

$$= Y / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]^{1/2} \sin(\omega t - \phi)$$

$$\phi = \text{Arctan}[2\xi(\omega/\omega_n) / (1 - (\omega/\omega_n)^2)]$$

Solution 2

$$m d^2x/dt^2 + c dx/dt + kx = c dy/dt = cY \omega \cos \omega t$$

doing derivatives leading to

$$X[(k - m\omega^2)\cos(\omega t - \phi) - c\omega \sin(\omega t - \phi)] = F \cos \omega t$$

using cosine sum leading to

Superposition method

$$\begin{aligned}X_2 &= (F/k) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2} \cos(\omega t - \phi) \\&= Y(c\omega/k) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2} \cos(\omega t - \phi) \\&= Y(2\xi(\omega/\omega_n)) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2} \cos(\omega t - \phi) \\\phi &= \text{Arctan}[2\xi(\omega/\omega_n) / (1 - (\omega/\omega_n)^2)]\end{aligned}$$

Total solution

$$\begin{aligned}X &= x_1 + x_2 = Y / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2} \sin(\omega t - \phi) + \\&\quad Y(2\xi(\omega/\omega_n)) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2} \cos(\omega t - \phi) \\&= Y' \sin(\omega t - \phi + \alpha) = Y' \sin(\omega t - \psi), \quad \text{where } \tan \alpha = 2\xi\omega/\omega_n \\Y' &= \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]} \\\tan(\phi - \alpha) &= (\tan \phi - \tan \alpha) / (1 + \tan \phi \tan \alpha) = 2\xi(\omega/\omega_n)^3 / (1 - (\omega/\omega_n)^2 + (2\xi(\omega/\omega_n))^2)\end{aligned}$$

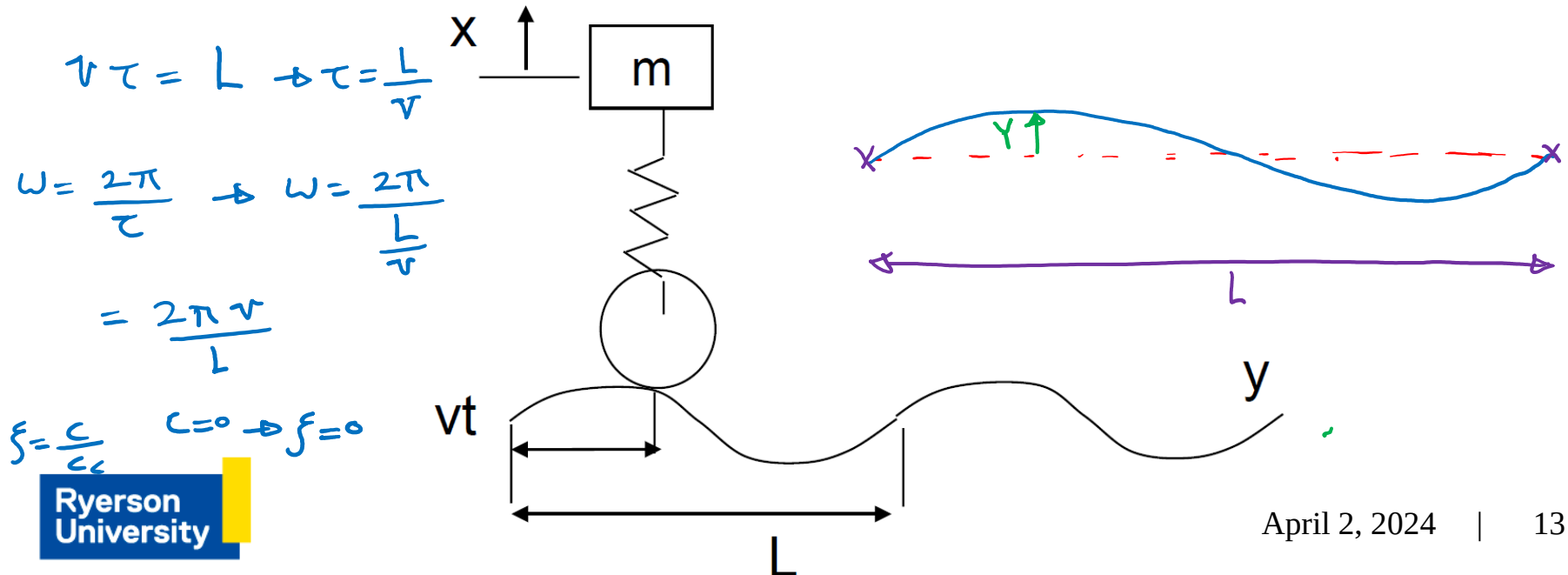
Example 10.1 wavy road

Problem:

The given system represents a car wheel supported by a suspension spring. The wheel is traveling on a bumpy road described by a sine wave.

Required:

1. Derive the equation of the amplitude X as a function of speed v .
2. Determine the most unfavorable speed.



Example 10.1 Cont.

Solution

$$|X/Y| = \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]}$$

→ slide #8

Since $c=0 \rightarrow \xi=0$, then

$$X/Y = 1/[1 - (\omega/\omega_n)^2] \rightarrow X = Y/[1 - (\omega/\omega_n)^2] \rightarrow X = \frac{Y}{1 - (\frac{\omega}{\omega_n})^2} = \frac{Y}{1 - (\frac{2\pi v}{L})^2}$$

$$\tan\psi = mc\omega^3 / [(k^2 - km\omega^2 + (c\omega)^2)] \rightarrow \psi = 0$$

Hence $x = X \sin \omega t$

Furthermore, the period is

$$1) \quad \tau = L/v = 2\pi/\omega \rightarrow \omega = 2\pi v/L \quad (L - \text{wave length})$$

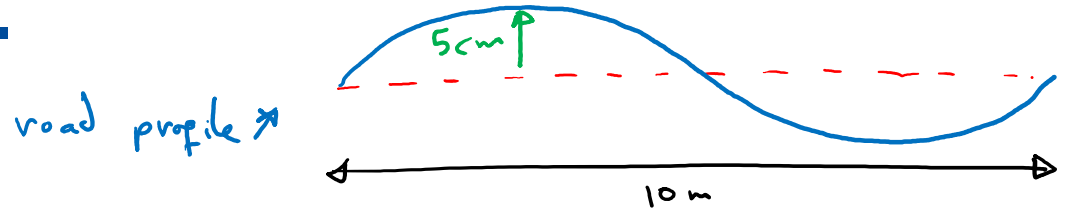
The most unfavorable speed corresponds to $\omega/\omega_n = 1$

$$\rightarrow \omega_n = (k/m)^{1/2} = 2\pi v/L$$

$$2) \quad v = L(k/m)^{1/2} / 2\pi$$

↓
most unfavorable speed

Example 10.1 Cont.



If $k=100$ N/m, $m=1$ kg, wave length $L= 10$,

The most unfavorable speed is

$$v=10(100/1)^{1/2}/2\pi=100/2\pi=15.9 \text{ m/s} \rightarrow \omega_n = (k/m)^{1/2}=10$$

What will be the amplitude of vibration at $v = 10$ m/s if $Y = 5$ cm?

Since $X=Y/[1-(\omega/\omega_n)^2]$

Then

$$\omega=2\pi v/L=2\pi 10/10=6.28$$

Hence

$$\omega/\omega_n=6.28/10=0.628$$

$$X=Y/[1-(\omega/\omega_n)^2]=5/(1-0.628^2)= 5 \times 1.65 = 8.25 \text{ cm}$$

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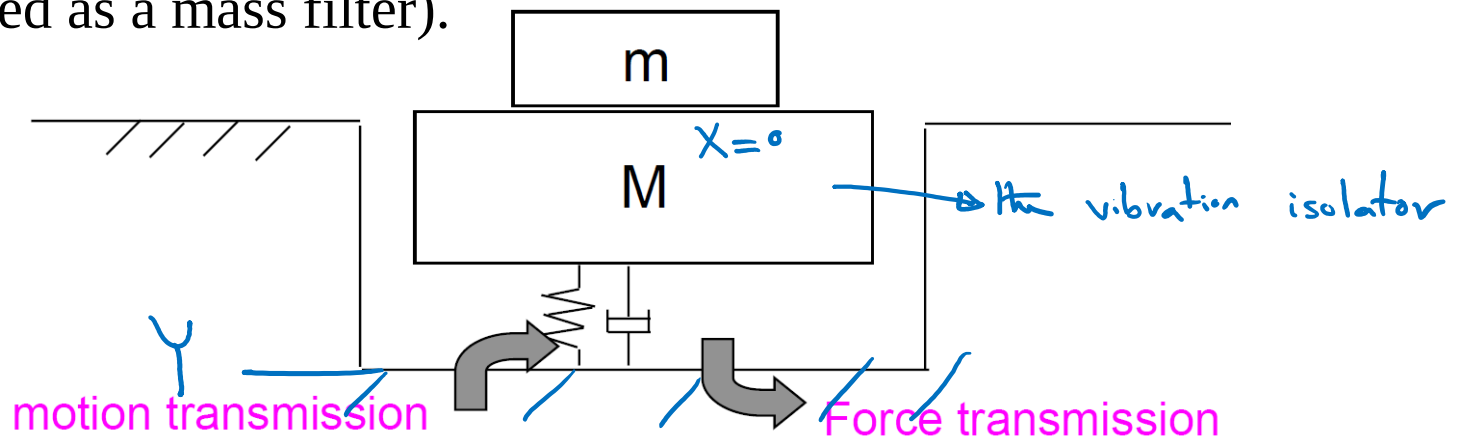
Text: Chapter 3

Vibration isolation

$$\text{if } \frac{\omega}{\omega_n} \rightarrow \infty \Rightarrow \left| \frac{x}{y} \right| \rightarrow 0 \Rightarrow x \rightarrow 0$$

$$\text{if } \frac{\omega}{\omega_n} \rightarrow \infty \rightarrow \omega_n \rightarrow 0 \quad \omega_n = \sqrt{\frac{k}{m}} \quad k \rightarrow \infty$$

- A vibration isolation system is used to protect a device from surrounding vibration excitation → **motion transmission isolation**
- It is also used to prevent the vibration generated by a machine from being transmitted to surrounding → **force transmission isolation**
- A typical vibration isolation system is a rotating machine mount, where the machine is mounted on a foundation supported by a spring-damper system.
- The mass of machine $\ll$ the mass of foundation (which can be considered as a mass filter).



Motion transmission isolation

- In motion transmission isolation, there is no internal vibration source. The excitation is the external support motion.
- Hence we need to look at the support motion vibration.

$$|X/Y| = \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]}$$

when $\omega/\omega_n \gg \sqrt{2}$, amplitude ratio < 1 , $X/Y = 1/(\omega/\omega_n)^2 \rightarrow 0$

- This means that the isolation system requires a small ω_n .
- A small ω_n can be realized by a soft spring and/or a relative heavy base.
- A typical example is an optical table which is made of marble supported by an air cushion system.
- Looking at the equation of motion

$$m d^2x/dt^2 = -k(x-y) - c(dx/dt - dy/dt)$$

$$\rightarrow m d^2x/dt^2 + c dx/dt + kx = \underline{ky + c dy/dt}$$

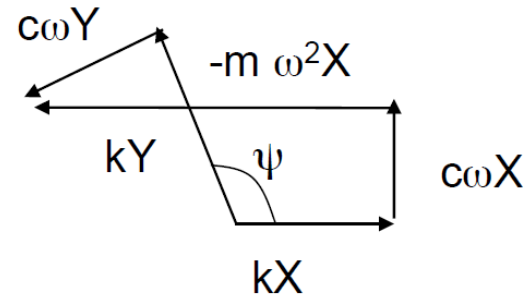
Equivalent transmitting force
through spring & damper

Motion transmission isolation

- $\omega/\omega_n \gg \sqrt{2}$, $\psi > 120^\circ$

$$X/Y = 1/(\omega/\omega_n)^2 \rightarrow 0$$

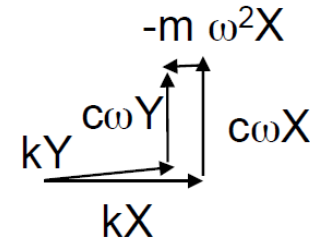
Transmitting force is used to overcome the inertia force.



- $\omega/\omega_n = 0$, $\psi = 0^\circ$

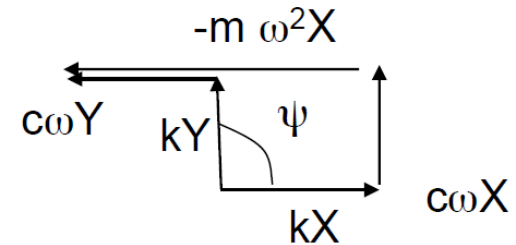
$$X/Y = 1$$

Transmitting force is used to overcome the spring and damping force.

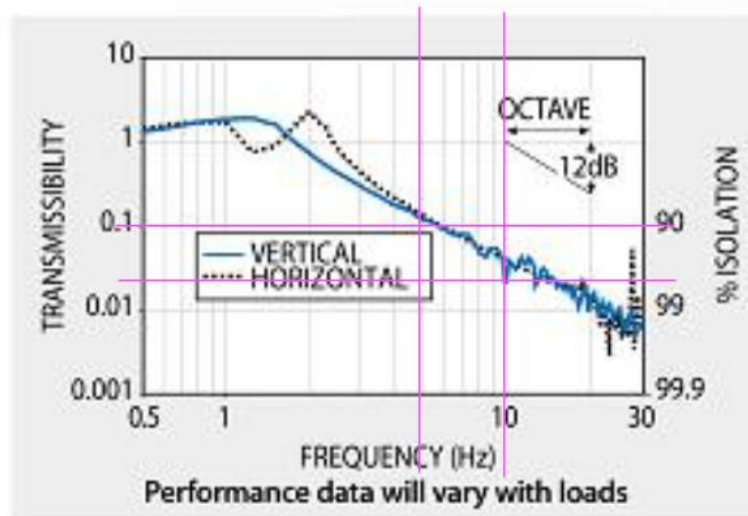


- $\omega/\omega_n = 1$, $\psi = 90^\circ$

$$X/Y = \sqrt{(1/2\xi)^2 + 1} \approx 1/2\xi$$



Optical table



Specifications

Isolation Efficiency

Vertical Natural Frequency	$\leq 1.1\text{Hz}$
Isolation Efficiency @ 5Hz	90%
Isolation Efficiency @ 10Hz	98.5%

Horizontal Natural Frequency	$\leq 1.9\text{Hz}$
Isolation Efficiency @ 5Hz	85%
Isolation Efficiency @ 10Hz	97%

Transmissibility & isolation efficiency

- Transmissibility

$$|X/Y| \rightarrow |\text{output amplitude} / \text{input amplitude}|$$

- Isolation efficiency

$$[(Y-X)/Y] \times 100\% \quad \frac{Y-X}{Y} \times 100\% = \left(1 - \frac{X}{Y}\right) 100\%$$

- Decibel (p.12)

$$\text{dB} = 20 \log (X/Y)$$

Example

$$|X/Y| = 0.1 \rightarrow [(1-0.1)/1]100\% = 90\%$$

$$\rightarrow 20 \log 0.1 = -20 \text{ dB}$$

$$|X/Y| = 0.015 \rightarrow [(1-0.015)/1]100\% = 98.5\%$$

$$\rightarrow 20 \log 0.015 = -36.5 \text{ dB}$$

← isolation efficiency
 $\log \rightarrow \log_{10}$ $\log_{10} 0.1 = \log_{10} 10^{-1} = -1$

Force transmission isolation

- In force transmission isolation, we look at the transmitting force from the spring-damper system to the support, such as imbalance induced vibration

kx ← spring force → damper force $c \frac{dx}{dt}$

Since the spring force and damping force are perpendicular, then the amplitude of the transmitting force is equal to

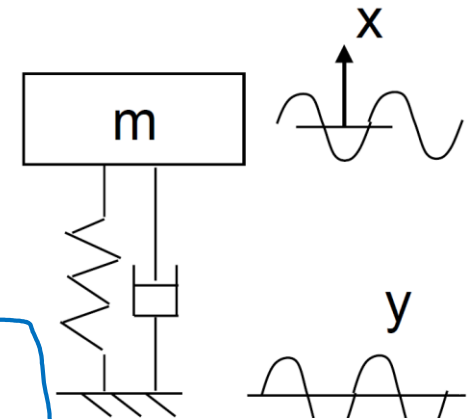
$$F_T = \sqrt{(kX)^2 + (c\omega X)^2} = kX[1 + (2\xi\omega/\omega_n)^2]^{1/2}$$

The force included vibration takes the form as

$$X = (F_O/k) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2}$$

The force transmissibility is defined as

$$F_T/F_O = \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]}$$



Transmissibility

- Both isolation cases deal with the same ratio, called transmissibility

$$TR = F_T / F_O \quad \text{In to out, force transmission.}$$

$$= X/Y \quad \text{Out to in, motion transmission.}$$

- When damping is negligible,

$$TR = 1 / [(\omega / \omega_n)^2 - 1] \quad \text{for } \omega / \omega_n > \sqrt{2}$$

- Recall $\omega_n = (g/\Delta)^{1/2}$, then $\rightarrow \Delta$ is deflection of the spring under the mass

$$TR = 1 / [(2\pi f)^2 \Delta / g - 1] \quad mg = k\Delta \rightarrow \Delta = \frac{mg}{k} \quad \omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{mg}{\Delta}} = \sqrt{\frac{g}{\Delta}}$$

- Hence the basic requirements for both cases are the same, i.e. using small ω_n .

- In addition, damping (air, liquid, friction) is used for vibration attenuation.

$$TR = \frac{1}{\left(\frac{\omega}{\omega_n}\right)^2 - 1} = \frac{1}{\frac{\omega^2}{g/\Delta} - 1}$$

$$TR = \frac{1}{\left(\frac{\omega}{\omega_n}\right)^2 - 1} \quad \begin{array}{l} TR \rightarrow 0 \\ \frac{\omega}{\omega_n} \rightarrow \infty \end{array} \quad \begin{array}{l} \omega \rightarrow \infty \\ \omega_n \rightarrow 0 \end{array}$$

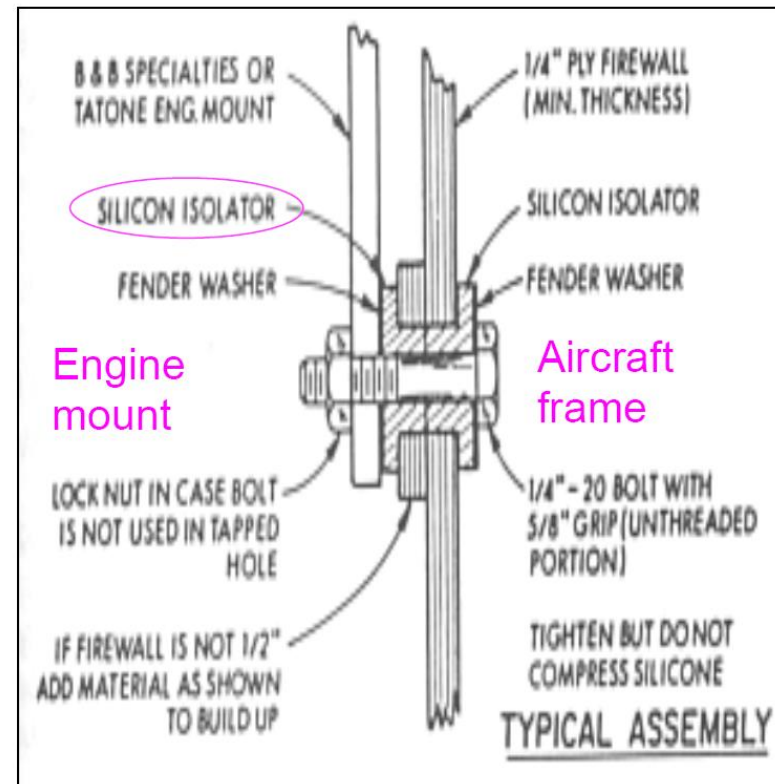
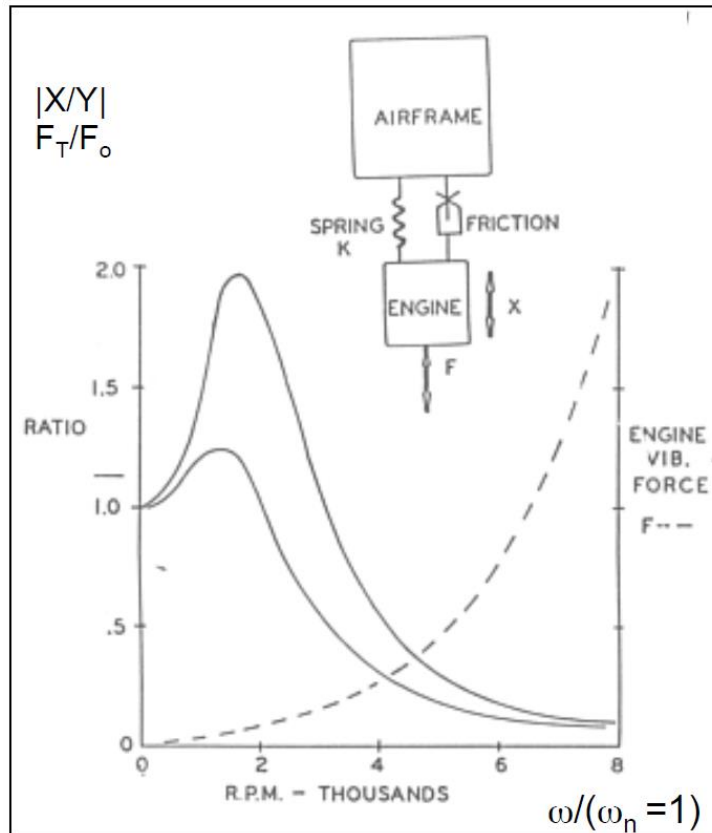
Aircraft engine mount

$$\text{unbalance force} = me\omega^2$$

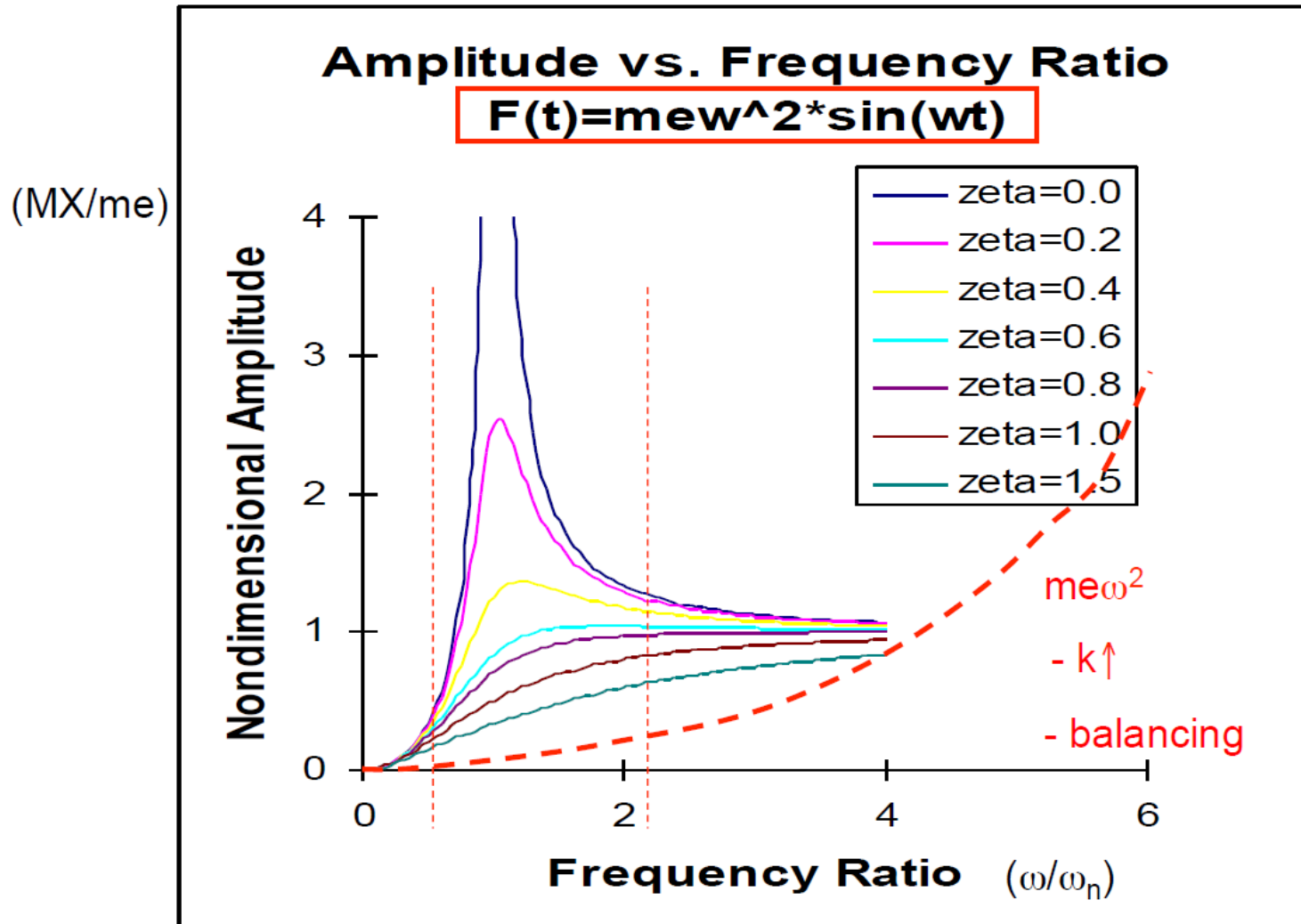
$$TR = \frac{F_T}{F_0} = \frac{F_T}{me\omega^2}$$

$$F_T = me\omega^2 \times TR$$

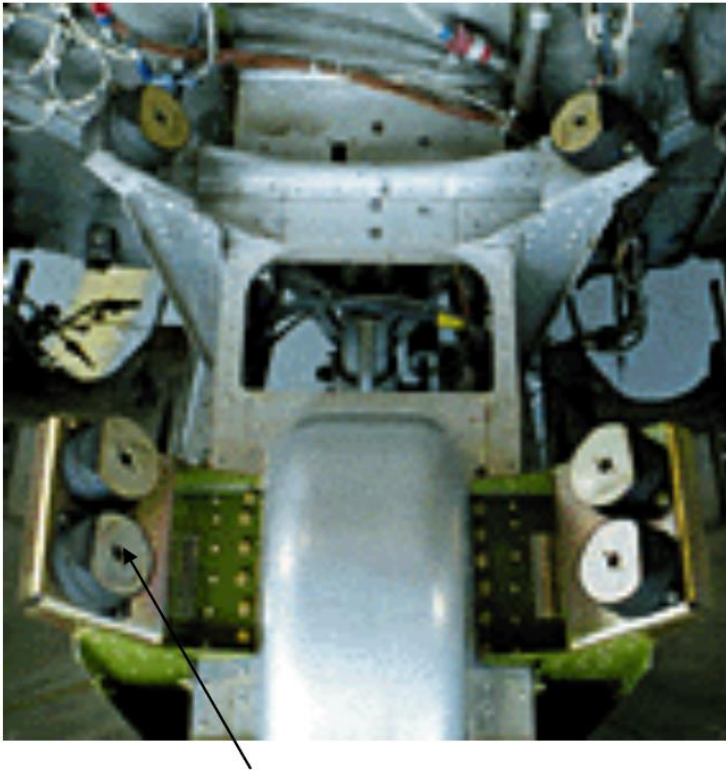
$(\omega/\omega_n) \uparrow$, $TR \downarrow$. However, $(\omega/\omega_n) \uparrow$, $MX/me \rightarrow 1$, balancing required



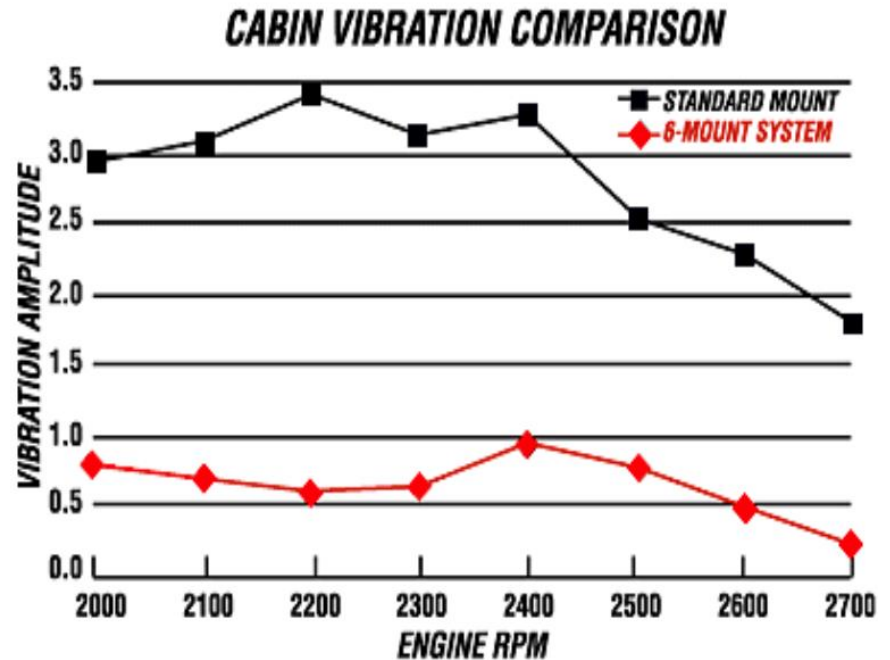
Imbalance induced vibration amplitude



Aircraft engine mount



Engine mount



Example 10.2 vibration isolation

$$F_0 = 350 \text{ N}$$

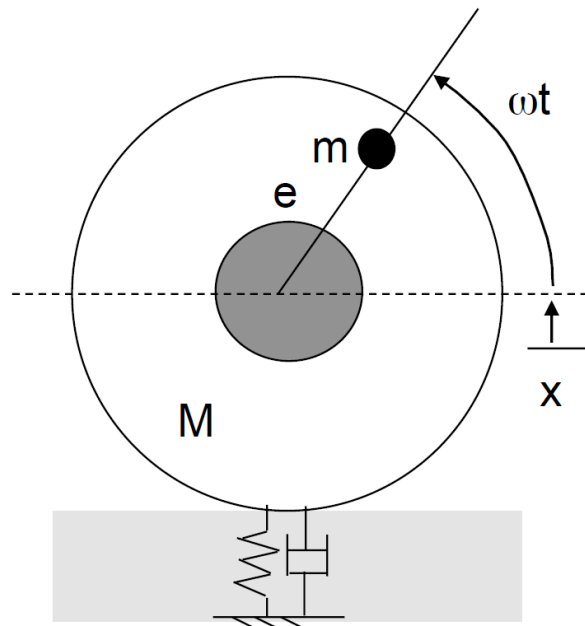
$$\omega = 3000 \text{ rpm}$$

$$\xi = 0.2$$

An engine of 100 kg is mounted on a spring of $k=700 \text{ kN/m}$. The unbalance induced vibration is 350 N at 3000 rpm, and $\xi=0.2$.

Required to determine:

1) Force induced vibration amplitude, 2) transmissibility, 3) transmitted force.



Example 10.2 vibration isolation

$$1) \quad \omega_n = (k/m)^{1/2} = (700 \times 1000 / 100)^{1/2} = 83.67 \text{ (rad/s)}$$

$$\omega = (3000/60) \times 2\pi = 100\pi \text{ (rad/s)}$$

week #9 $\Rightarrow$

$$X = (F_O/k) / [(1 - (\omega/\omega_n)^2)^2 + (2\xi(\omega/\omega_n))^2]^{1/2}$$

$$= (350/700 \times 1000) / [(1 - (100\pi/83.67)^2)^2 + (2 \times 0.2 \times 100\pi/83.67)^2]^{1/2}$$

$$= 0.0379 \text{ mm}$$

$$2) \quad TR = \sqrt{[1 + (2\xi\omega/\omega_n)^2] / [(1 - (\omega/\omega_n)^2)^2 + (2\xi\omega/\omega_n)^2]}$$

$$= \sqrt{[1 + (2 \times 0.2 \times 100\pi/83.67)^2] / [(1 - (100\pi/83.67)^2)^2 + (2 \times 0.2 \times 100\pi/83.67)^2]}$$

$$= 0.137$$

$$3) \quad F_T = F_O \times TR = 350 \times 0.137 = 47.89 \text{ N}$$

What is the base vibration? $\leftarrow$

week #10

Ryerson
University

$$| \frac{X}{Y} | = \sqrt{ \frac{1 + (2\xi \frac{\omega}{\omega_n})^2}{(1 - (\frac{\omega}{\omega_n})^2)^2 + (2\xi \frac{\omega}{\omega_n})^2} }$$

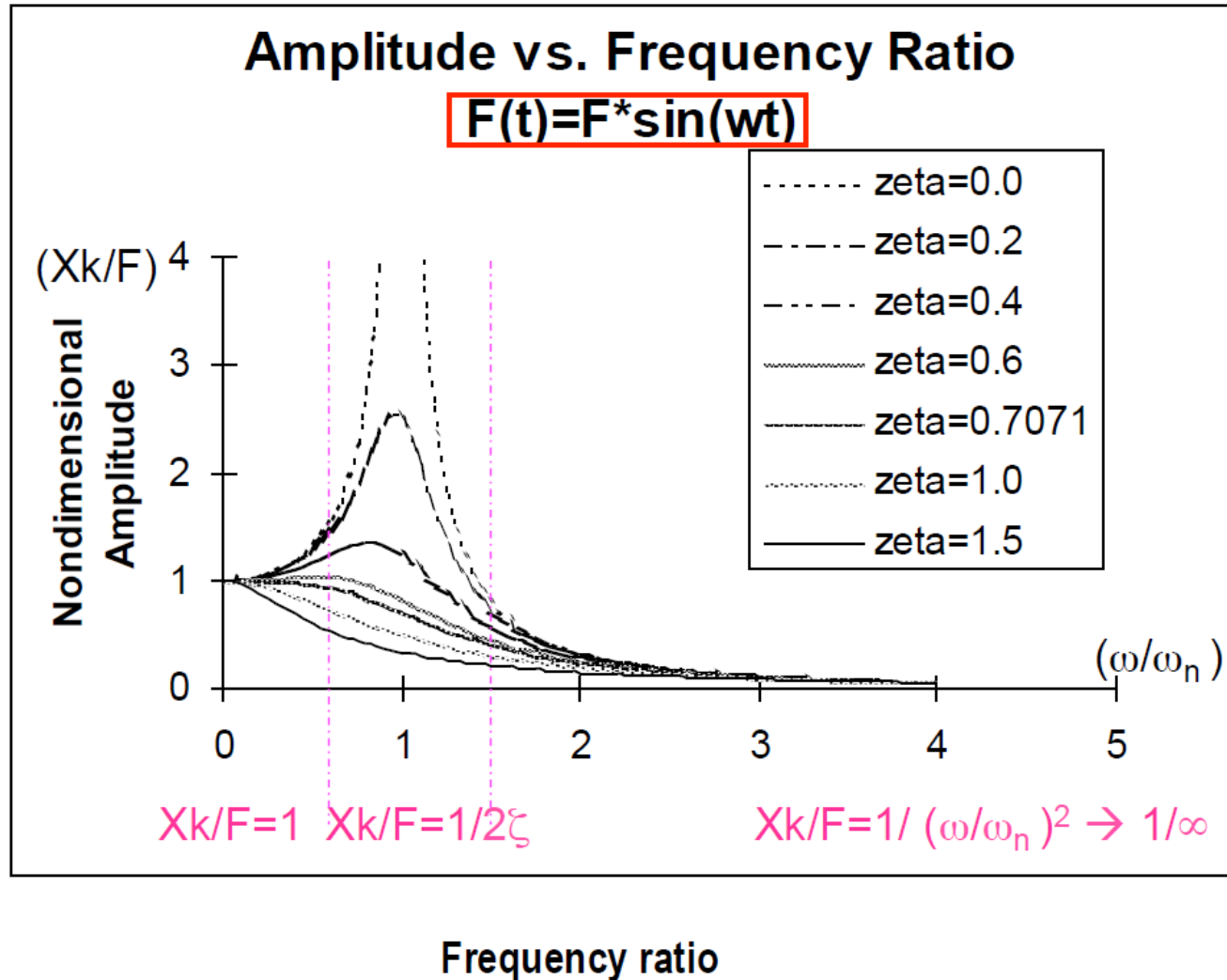
$\omega, \omega_n, \xi \checkmark$

$$\xrightarrow{\quad} X = 0.0379 \text{ v}$$

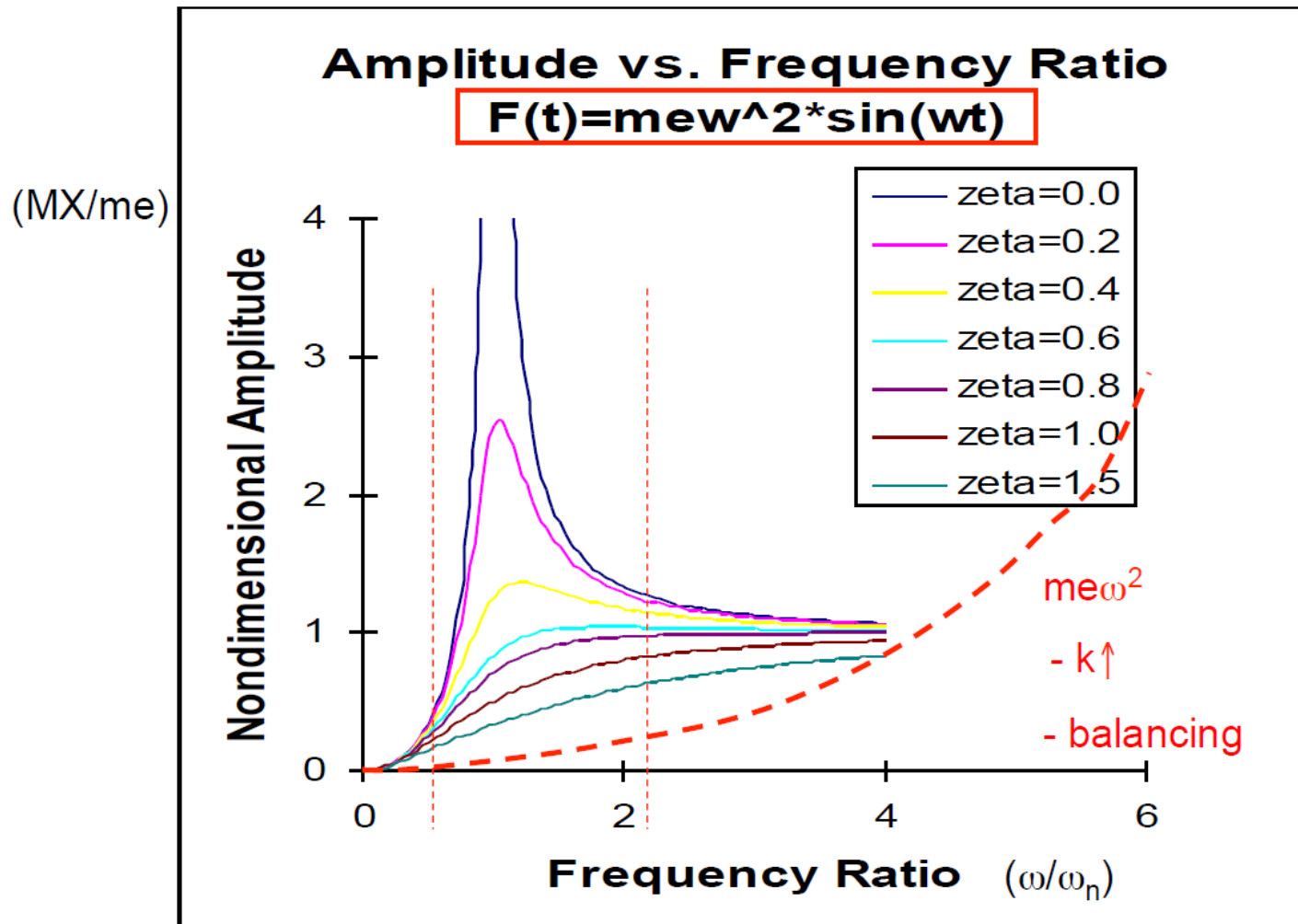
$Y \rightarrow$ can be found

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vibration of k support

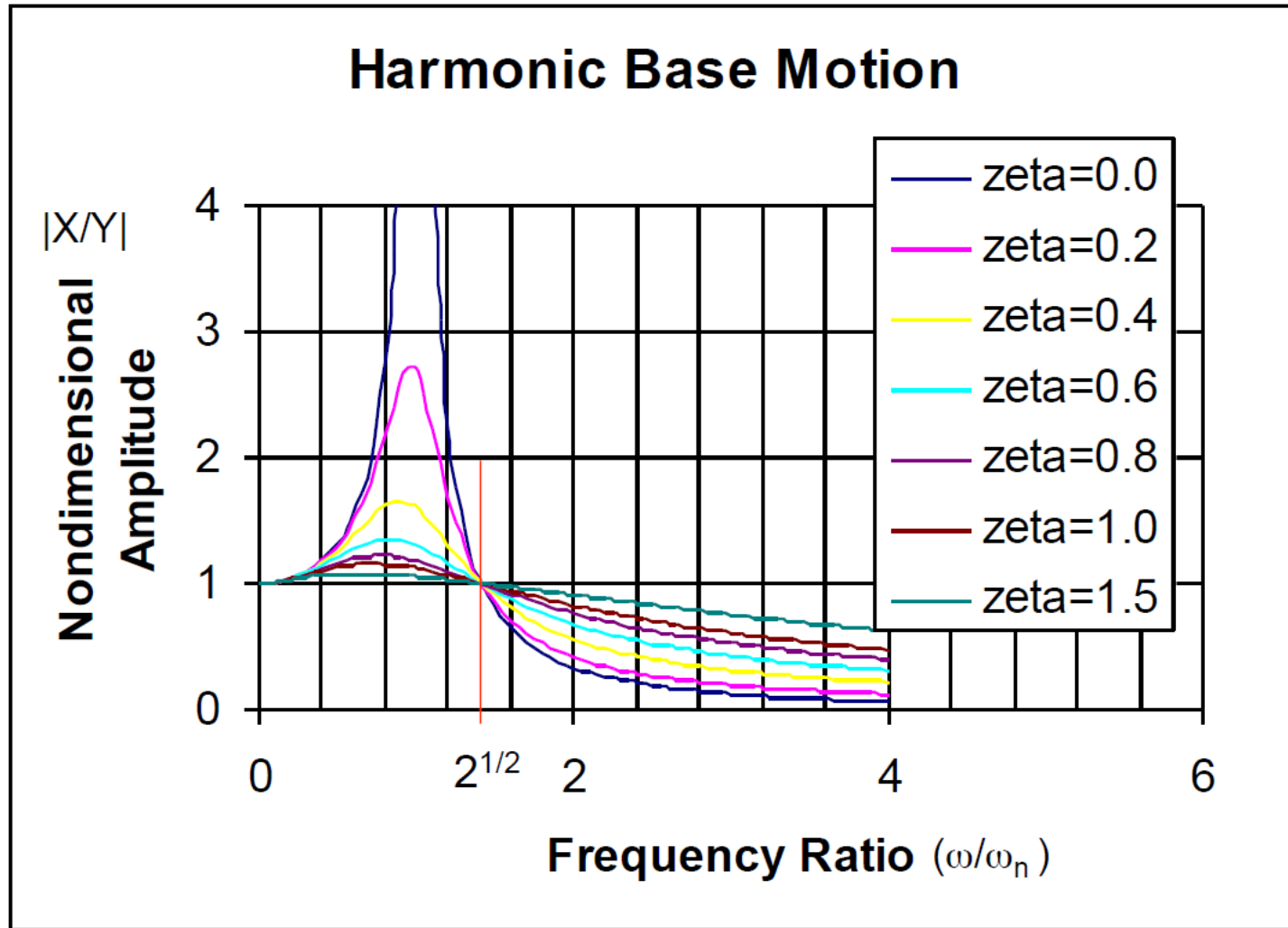
Forced vibration amplitude



Imbalance induced vibration amplitude



Amplitude vs frequency ratio



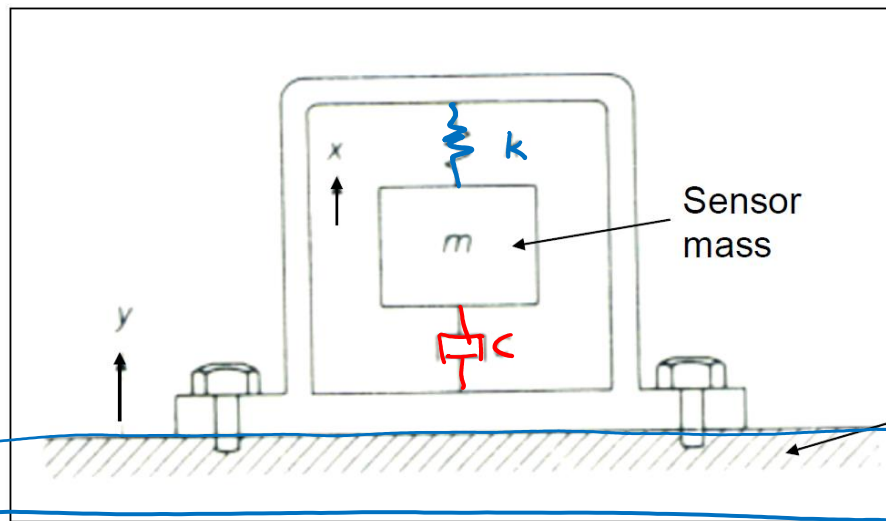
Course outline

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- Vibration sensors

Text: Chapter 3

Vibration sensors

- Vibration sensors are used to measure vibrations.
- The principle of the vibration sensors is based on support motion.
- The vibration measured is the vibration of the mass inside the sensor.
- Therefore, the vibration measured is the relative motion between the sensor and the vibrating body, i.e. $z=x-y$.

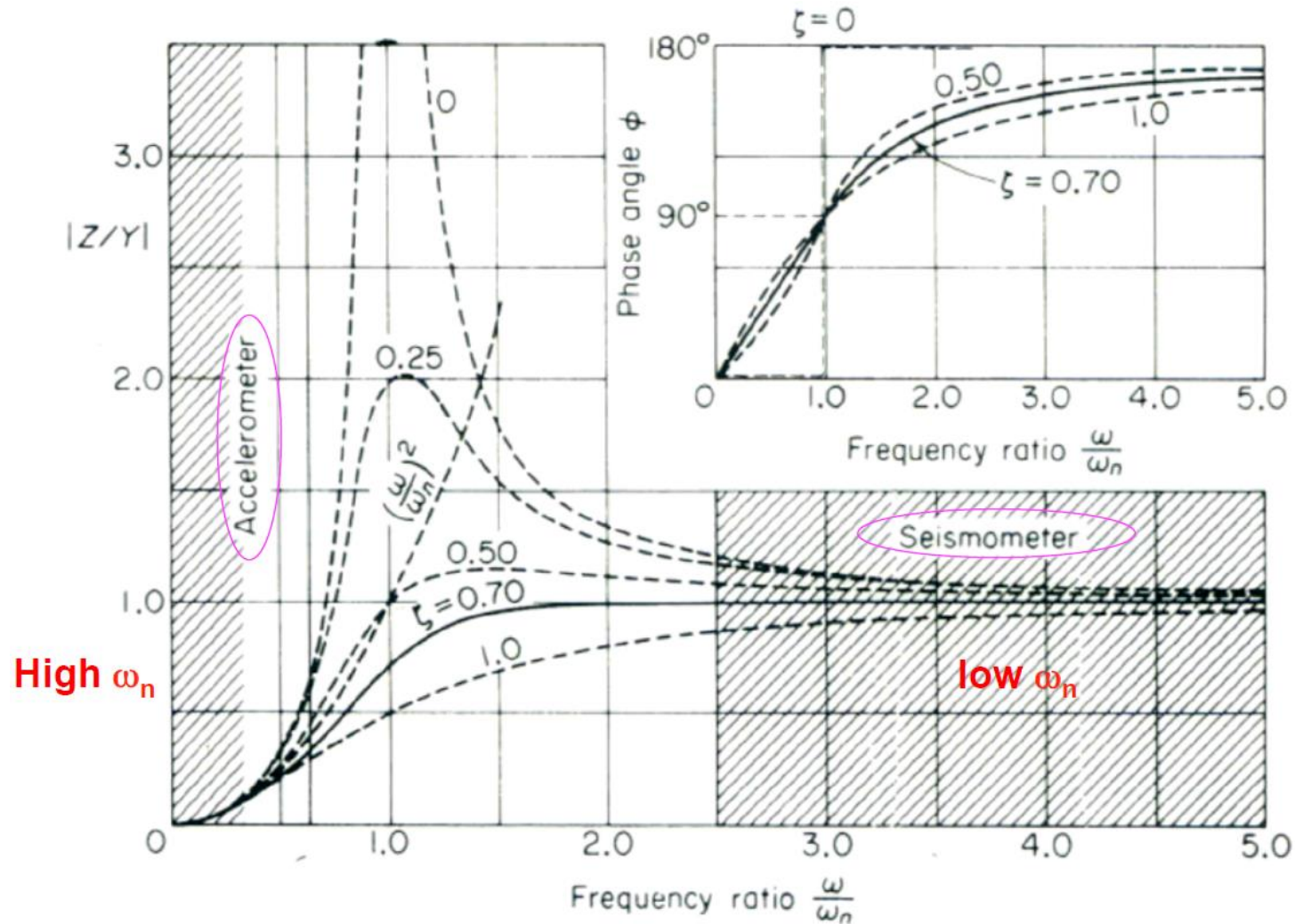


vibrat lab #1
attach sensor to the
cantilever beam, y is needed
(support motion)

x is measured based on
Vibrating
body ~~v~~el~~a~~tage

$$\left| \frac{x}{y} \right| = \sqrt{\quad} \rightarrow Y \sqrt{\quad}$$

Relative vibration measurement



Accelerometers

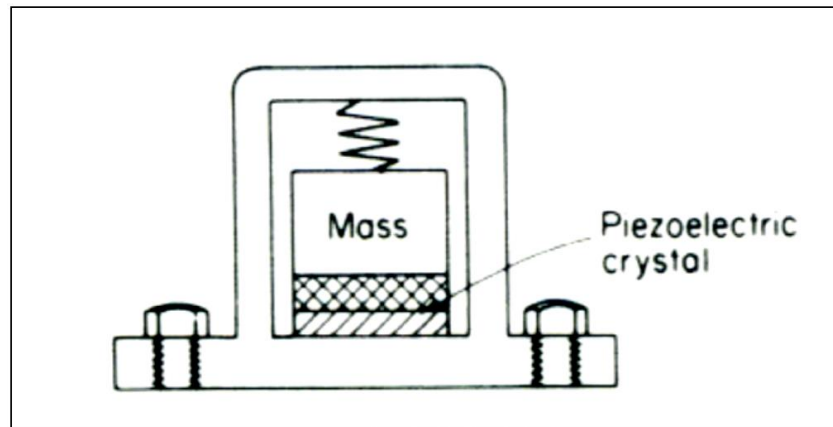
$$Z = \left(\frac{\omega}{\omega_n} \right)^2 Y$$

$$\begin{aligned} y &= Y \sin \omega t \\ \dot{y} &= Y \omega \cos \omega t \\ \ddot{y} &= -Y \omega^2 \sin \omega t \end{aligned}$$

↙ acceleration

- Accelerometers are designed utilizing support motion in the range of $\omega / \omega_n \approx 0$. $\frac{\omega}{\omega_n} \ll 1 \rightarrow \omega_n \rightarrow \infty$
- This means that the natural frequency of the sensor is much higher than the vibration frequencies being measured.
- In this case, the vibration sensed by the accelerometer is proportional to the acceleration of the vibrating body.

$$Z = (\omega / \omega_n)^2 Y = \omega^2 Y / \omega_n^2 = \text{acceleration} / \omega_n^2$$



Steady-state solution

$$z = x - y$$

The steady-state solution is

$$z = Z \sin(\omega t - \phi)$$

where

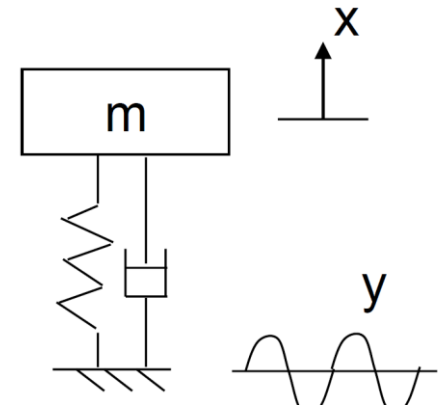
$$\text{Amplitude: } Z = m\omega^2 Y / [(k - m\omega^2)^2 + (c\omega)^2]^{1/2}$$

$$\text{Phase: } \phi = \text{Arctan}[c\omega / (k - m\omega^2)]$$

If $\omega \ll \omega_n$ then

$$Z \approx m\omega^2 Y / k = (\omega / \omega_n)^2 Y$$

$$\phi \approx 0$$



Accelerometers

we have an accelerometer of $5 \frac{mV}{g}$ if it reads 10.06 V, what is the acceleration

- Accelerometers are made of piezoelectric crystals that have high natural frequency in the range of over 50 KHz. $\rightarrow$ very high natural frequency
- The size of the accelerometer is small, about 1 cm, and very light between 0.5-50 grams.
- Accelerometers measure the acceleration in range of 100 – 10k (g).
- The sensitivity of the accelerometer is given in mV/g.

Example: if $mV/g=5$, vibration is $y=2\sin 1000\pi t$ (mm), what will be the output voltage?

$$y = Y \sin \omega t \quad \rightarrow \quad \begin{aligned} \omega &= 1000\pi \\ Y &= 2 \text{ mm} \end{aligned}$$

$$A = (\omega^2 Y / g) \times g = ((1000\pi)^2 \times 0.002 / g) \times g = 2012 \text{ g}$$

$$\text{Volt} = 2012 \text{ g} \times 0.005 \text{ V/g} = 10.06 \text{ Volts}$$

$$\text{acceleration} = Y \omega^2 = \frac{0.002 \times (1000\pi)^2}{g} = \left(\frac{0.002 \times 1000 \times 1000 \pi^2}{g} \right) = 2012 \text{ g}$$

$$5 \frac{mV}{g}$$

$$\text{output voltage} = 2012 \text{ g} \times 5 \times 10^{-3} \frac{V}{g} = 10.06 \text{ V}$$

Velocity pick-up

- Velocity pick-ups are designed utilizing support motion in the range of $\omega/\omega_n \gg 1$ $\frac{\omega}{\omega_n} \gg 1$ $\omega_n \rightarrow 0$
- This means that the natural frequency of the sensor is much lower than the vibration frequencies to be measured.
- In this case, the vibration sensed by the velocity pick-ups is proportional to the displacement of the vibrating body.
 $Z=Y$
- Since the relative motion Z is measured by solenoid (magnet), which is proportional to the velocity of the sensor mass, it is called velocity pick-up.
- The sensitivity of the velocity pick-up is given as mV/(cm/s) (20-350). The natural frequency of the sensor itself is 1-5 Hz, and the measurable frequency is between 10-2000 Hz, with peak limit of 5 mm.

Velocity pick-up

Example: $\text{mV}/(\text{cm/s}) = 20$, vibration is $y = 2\sin 1000\pi t$ (mm), what will be the output voltage?

$$y = Y \sin \omega t$$

$$v = \omega Z = \omega Y = 1000\pi \times 0.2 = 628 \text{ cm/s}$$

$$Y = 2 \text{ mm}$$

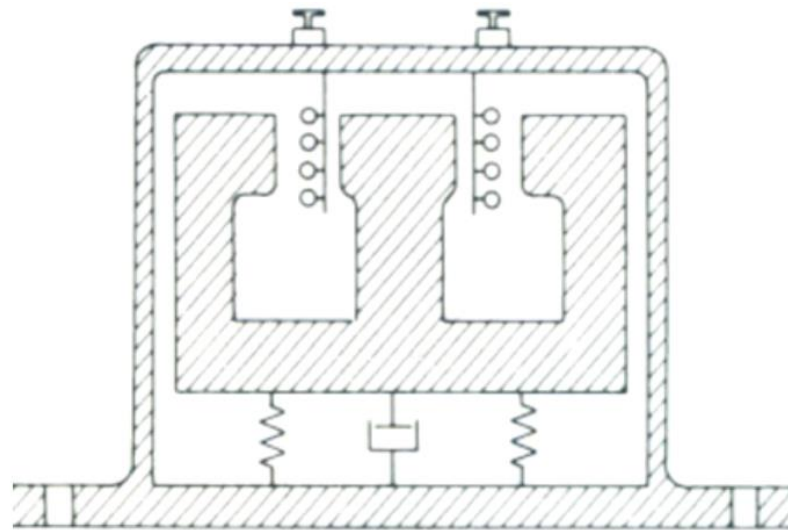
$$\omega = 1000\pi$$

$$\text{Volt} = 628 \times 20 / 1000 = 12.56 \text{ volts}$$

$$20 \frac{\text{mV}}{\text{cm/s}}$$

output voltage =

$$20 \frac{\text{mV}}{\text{cm/s}} \times 628 \frac{\text{cm}}{\text{s}}$$



Vibration assignment #3: support motion

Problems 3.21, 3.23, 3.28